

I want to reuse playerId and salary-related fields. Should Salary extend Player in a Java football club management system, or should Salary be a separate record linked through playerId? Explain the correct OOP relationship.

Salary should not extend Player.

Inheritance means an "is-a" relationship:

- StarPlayer is a Player
- RegularPlayer is a Player
- Salary is not a Player

A Salary is a record about a Player's payment for a specific month, so the correct relationship is association:

Player 1 ----- many Salary records
linked by playerId

Keep Salary as a separate class:

```
// Java

public class Salary extends ClubRecord {
    private String playerId;
    private int month;
    private int year;
    private double playerMonthlySalary;
    private double performanceBonus;
    private double totalSalary;
}
```

Player should contain player profile data and behavior.

`Player` should contain player-profile data and behavior:

```
Java

public abstract class Player extends ClubEntity {
    private String fullName;
    private int age;
    private String nationality;
    private String position;
    private int shirtNumber;
    private double baseSalary;
    private String status;

    public abstract double calculateMonthlySalary();
}
```

If `Salary` extends `Player`, every `Salary` object would inherit fields such as `fullName`, `age`, `nationality`, `position`, and `shirtNumber`. Those fields do not belong to a payment record and would create unnecessary duplication.

To reuse salary-related data correctly, use the `Player` object only when creating or updating a `Salary` record:

```
Java

Player player = playerlist.findPlayerById(salary.getPlayerId());

if (player != null) {
    salary.setPlayerMonthlySalary(player.calculateMonthlySalary());
}
```

Then `Salary` stores the calculated amount for that specific salary record. This keeps the class design clear:

- `Player` = current player profile and salary calculation rule.
- `Salary` = one monthly payment record linked to that player through `playerId`.

📄 🍏 🗨️ ↻ ...

```
public abstract class Player extends ClubEntity {
    ...
}

public abstract class ClubRecord extends ClubEntity {
    ...
}

public class Salary extends ClubRecord {
    ...
}
```